SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:		**Duration:**	60 minutes

Code of Conduct

I P. MUNI KARTHIK / 192425061 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ___________________________

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**

   - **Explain the problem in your own words.**

A Smart Warehouse Management System is a software application designed to automate warehouse operations such as inventory tracking, product storage, stock updates, order processing, and shipment management. Traditional warehouse systems rely heavily on manual record-keeping or spreadsheets, which often result in stock mismatches, misplaced products, delayed deliveries, and human errors. The proposed system digitizes warehouse activities, providing real-time inventory monitoring and improving operational efficiency.

- **Identify the objectives of the proposed system.**

The objectives of the Smart Warehouse Management System are:

- To automate warehouse operations and reduce manual work.
- To maintain accurate inventory records in real time.
- To improve product tracking from receiving to dispatch.
- To reduce stock shortages and overstocking.
- To speed up order processing and shipment.
- To provide reports and analytics for better decision-making.

- **State the scope of the problem.**

The system covers inventory management, product receiving, storage management, supplier management, order processing, shipment tracking, report generation, user management, and stock monitoring. It can be used in retail warehouses, manufacturing industries, logistics companies, and e-commerce distribution centers.

2. **Identify Key Stakeholders and Challenges**

- **Identify all the stakeholders involved in the system.**

The stakeholders include:

- Warehouse Manager
- Warehouse Staff
- Suppliers
- Customers
- System Administrator
- Company Management

- **Explain the roles and responsibilities of each stakeholder.**

Warehouse Manager

- Monitors warehouse activities.
- Approves stock movements.
- Generates reports.
- Manages warehouse performance.

Warehouse Staff

- Receives products.
- Stores products in assigned locations.
- Updates inventory.
- Picks and dispatches customer orders.

Suppliers

- Supply products to the warehouse.
- Update delivery information.
- Coordinate purchase orders.

Customers

- Place product orders.
- Receive deliveries.
- Track order status.

System Administrator

- Creates user accounts.
- Manages system security.
- Performs software maintenance.
- Maintains database backups.

Company Management

- Reviews warehouse performance.
- Makes business decisions using reports.
- Approves operational policies.

    - **Discuss the major challenges and issues faced by the stakeholders.**

- Warehouse managers struggle with inaccurate inventory information.
- Warehouse staff spend significant time updating records manually.
- Suppliers experience delays due to poor communication.
- Customers may receive delayed or incorrect orders.
- Administrators must ensure continuous system availability and security.
- Management lacks real-time insights for strategic decision-making.

3. **Analyze the Current Approach**

- **Describe the existing system or current process.**

The current warehouse system mainly uses paper records or spreadsheet software to record inventory transactions. Employees manually update stock when products are received or dispatched. Inventory checking is performed periodically instead of continuously.

- **Identify its strengths and limitations.**

Strengths

- Easy to implement.
- Low initial investment.
- Suitable for very small warehouses.
- Minimal technical infrastructure required.

Limitations

- High possibility of human errors.
- Time-consuming inventory updates.
- No real-time stock information.
- Difficult to locate products quickly.
- Limited reporting capabilities.
- Poor scalability for growing businesses.

- **Analyze the problems or gaps that need to be addressed.**

- Inventory mismatches.
- Duplicate stock records.
- Delayed order processing.
- Poor warehouse space utilization.
- Lack of real-time inventory visibility.

4. **Propose a Suitable Solution Using Software Engineering Principles**

- **Design a software-based solution for the given problem.**

Develop a centralized Smart Warehouse Management System that allows users to manage inventory, suppliers, customers, orders, warehouse locations, and reports through a web or mobile interface. Barcode or QR code scanning can be used for faster product identification and real-time inventory updates.

- **Describe the major functional modules of the proposed system.**

User Management Module

- User login
- Role-based access
- Password management

Inventory Management Module

- Add products
- Update products
- Delete products
- Stock monitoring

Receiving Module

- Record incoming goods
- Verify supplier deliveries
- Update inventory automatically

Storage Management Module

- Assign storage locations
- Track product locations

Order Management Module

- Receive customer orders
- Allocate inventory
- Generate invoices

Dispatch Module

- Prepare shipments
- Update dispatched stock
- Generate delivery details

Supplier Management Module

- Supplier registration
- Purchase order management
- Delivery tracking

Reporting Module

- Inventory reports
- Sales reports
- Stock movement reports
- Low-stock reports

  - **Identify the functional and non-functional requirements.**

- User authentication.
- Product management.
- Inventory tracking.
- Supplier management.
- Customer order management.
- Barcode scanning.
- Report generation.
- Notification for low stock.
- Search products.

  - **Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.**

- **Modularity**
  The system is divided into separate modules such as inventory, suppliers, orders, reports, and users, making development and maintenance easier.
- **Scalability**
  The system can easily support additional warehouses, users, and products without changing the existing architecture.

- **Maintainability**
  Well-structured modules and documented code allow developers to fix bugs and add new features efficiently.
- **Reliability**
  Automatic data validation, regular backups, and transaction logging ensure accurate and dependable warehouse operations.
- **Usability**
  A simple dashboard, intuitive navigation, and barcode scanning make the system easy to use for warehouse employees.

5. **Justify the Chosen Methodology/Framework**

   - **Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).**

Selected SDLC Model: Agile (Scrum)

   - **Justify why the selected methodology/framework is suitable for the proposed system.**

Agile is appropriate because warehouse requirements frequently change based on business growth and customer needs. Agile allows continuous customer feedback, incremental feature development, regular testing, and quick adaptation to changing requirements.

   - **Explain how it supports successful project development and maintenance.**

- Delivers working software in small iterations.
- Detects defects early through continuous testing.
- Encourages regular stakeholder feedback.
- Makes future enhancements easier.
- Reduces development risks.

6. **Discuss Expected Outcomes and Limitations**
   - **Describe the expected benefits of the proposed solution.**

- Accurate inventory records.
- Reduced manual errors.
- Faster warehouse operations.
- Improved order processing.
- Better warehouse space utilization.
- Real-time inventory visibility.

- **Explain how the solution addresses the identified challenges.**

The system automates inventory updates, reduces human errors, provides real-time stock information, improves communication among stakeholders, speeds up order processing, and generates accurate reports for management.

- **Discuss potential risks, implementation challenges, and limitations.**

Potential Risks

- Cybersecurity threats.
- Data loss due to hardware failure.
- Network connectivity issues.

Implementation Challenges

- Employee training.
- Initial deployment cost.
- Data migration from old systems.
- Integration with existing software.

Limitations

- Dependence on internet connectivity.
- Regular maintenance is required.
- High initial investment for barcode scanners and hardware.

- **Suggest possible future enhancements.**

- AI-based inventory demand forecasting.
- IoT sensors for smart inventory monitoring.
- RFID-based automatic product tracking.
- Mobile application for warehouse staff.
- Cloud-based warehouse synchronization.
- Predictive analytics for inventory optimization.
- Robot-assisted picking and packing.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/framework and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding	/5
Stakeholder & Challenge Analysis	/5
Current System Analysis	/5
Proposed Software Solution & Software Engineering Principles	/5
Methodology/ Framework Justification	/5
TOTAL	/5